CSC 6302 - Computer Vision

# Practical 6 Report

Motion, Tracking, Depth and Augmented Reality

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Target length: 4-6 pages

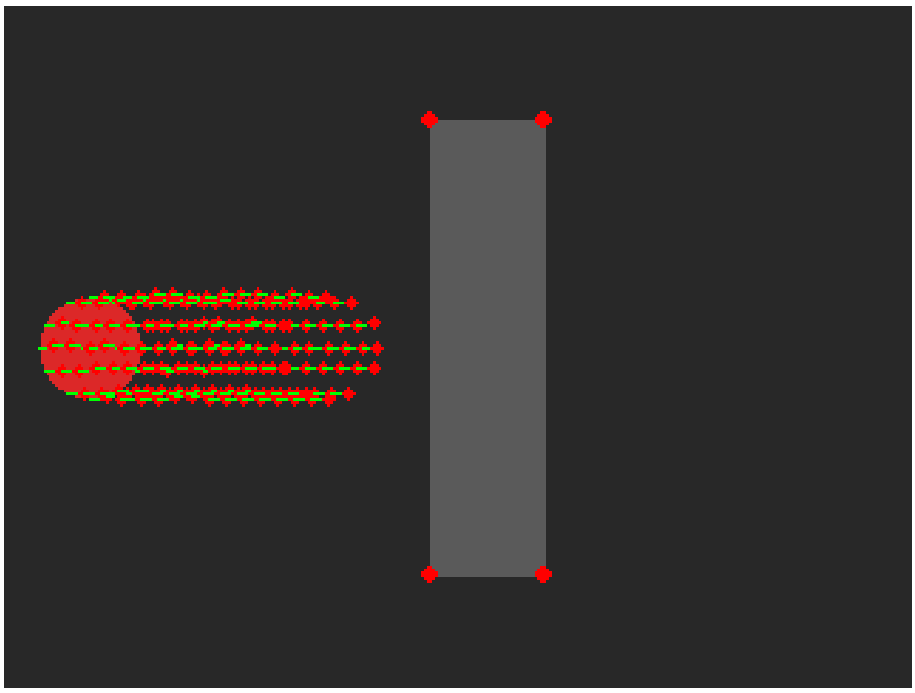
## 1. Overview

What I did: sparse LK + dense Farneback flow, my own Kalman filter through occlusion, chessboard calibration, AR cube with solvePnP, and stereo disparity. Demo video is recorded separately for submission.

## 2. Task 1 - Optical flow (4 marks)

LK tracks Shi-Tomasi corners with motion trails. Farneback gives dense flow; I visualise hue=direction, value=magnitude. Brightness-constancy fails when I suddenly brighten a frame - flow invents motion that is really just lighting. That is exactly the LK assumption from lectures.

### Figure 1. LK trails



### Figure 2. Farneback HSV flow

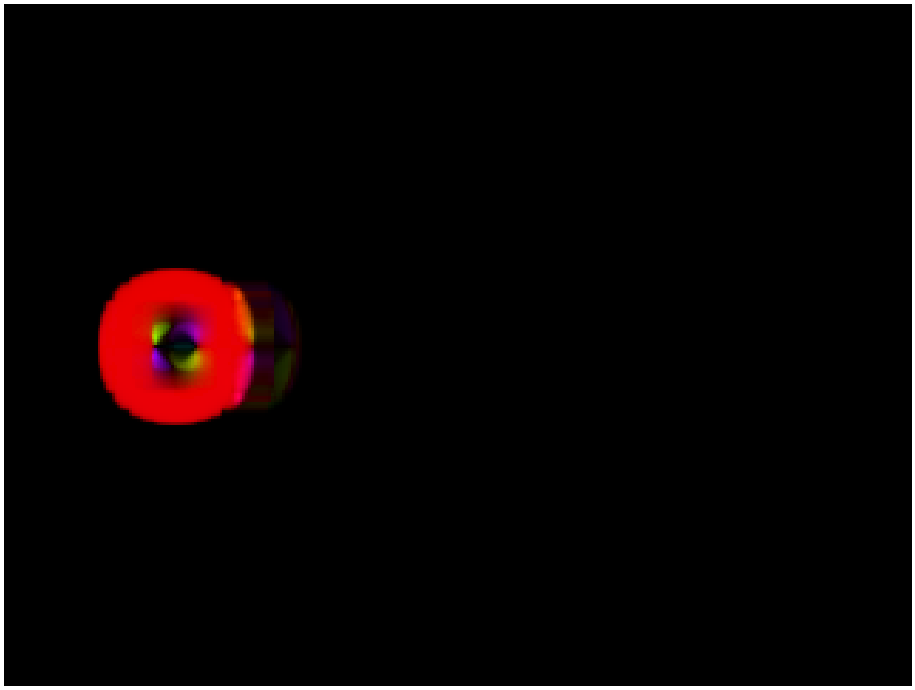
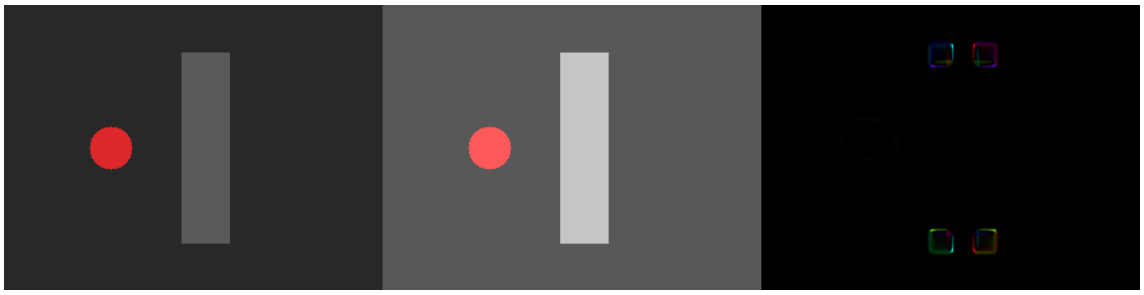


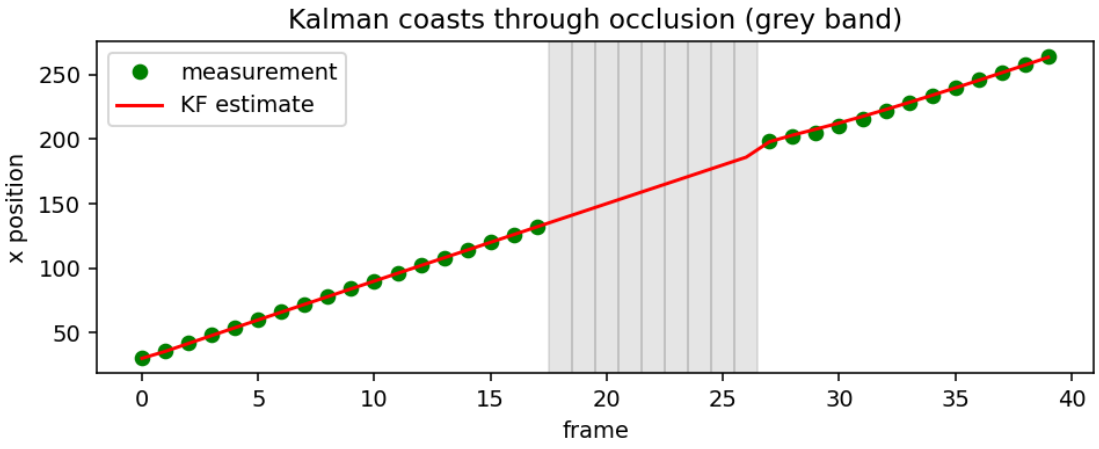
**Figure 3. Flow failure under lighting change**



## 3. Task 2 - Kalman tracking (5 marks)

What I did: HSV detect red object centroid, feed into my constant-velocity KF (predict/update written out). During occlusion z=None and the filter coasts. Tune: q=5e-2 r=8.0 (higher r = trust detector less). CSRT note: CSRT unavailable: module 'cv2' has no attribute 'TrackerCSRT_create'. CSRT/KCF often lose the lock when the object disappears; KF can keep predicting if motion is smooth.

**Figure 4. Measurement vs KF (grey = occlusion)**

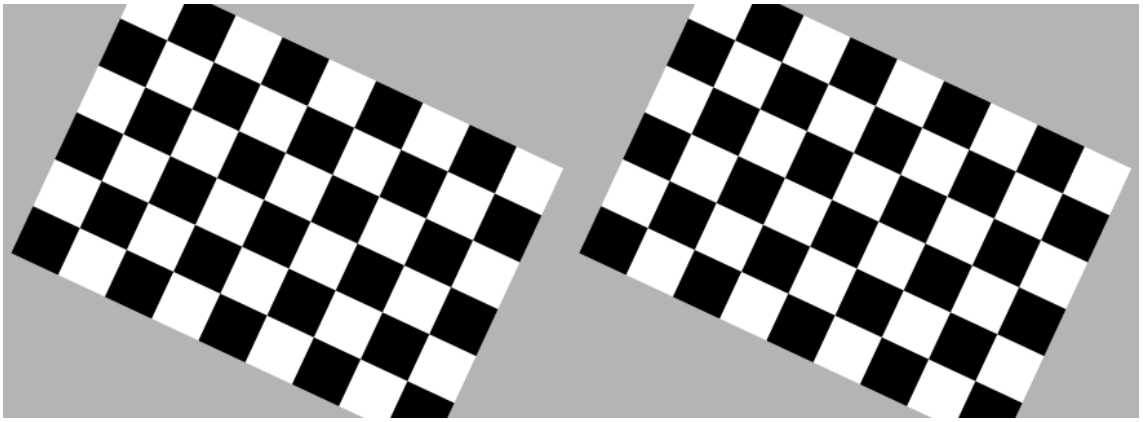



## 4. Task 3 - Calibration (4 marks)

What I did: findChessboardCorners + calibrateCamera on multiple views. `rms_reprojection_px=0.0371`

```
camera_matrix=
[[3.58014242e+04 0.00000000e+00 2.71365809e+02]
[0.00000000e+00 3.57988953e+04 1.28454355e+02]
[0.00000000e+00 0.00000000e+00 1.00000000e+00]]
dist_coeffs=[ 3.14012074e-01 6.00409921e-05 -3.06169676e-03 2.50142641e-03
1.57646912e-09]
fx_pixels=35801.42
views_used=11
```

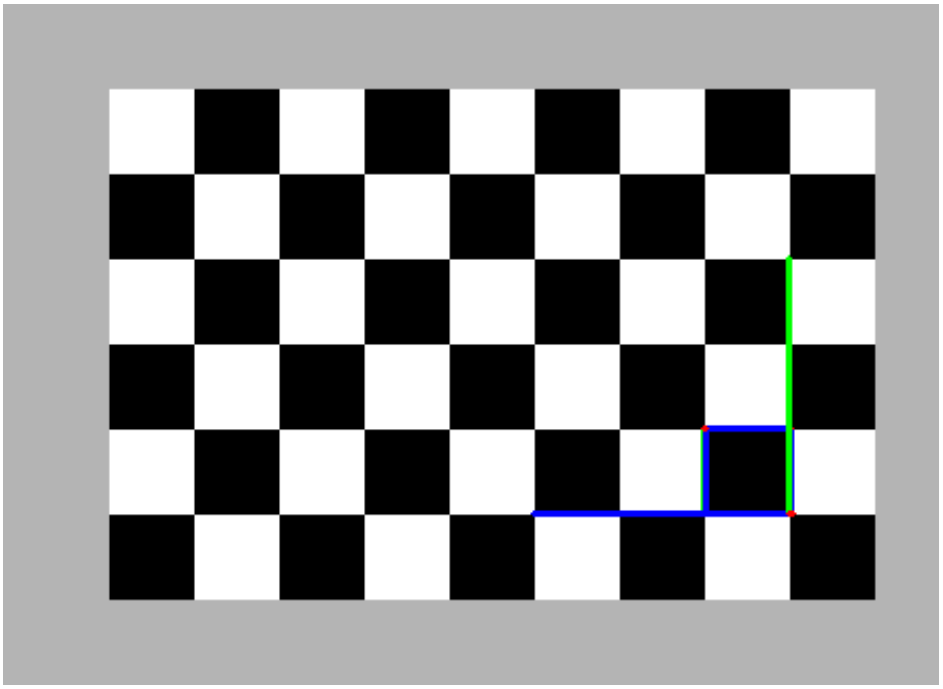
**Figure 5. Before / after undistort**



## 5. Task 4 - AR cube (4 marks)

solvePnP estimates camera pose (rvec, tvec) given known 3D board points and 2D detections plus intrinsics. Calibration had to come first because projection needs K (and distortion). I project cube vertices with projectPoints and draw edges + XYZ axes.

**Figure 6. AR cube on chessboard**

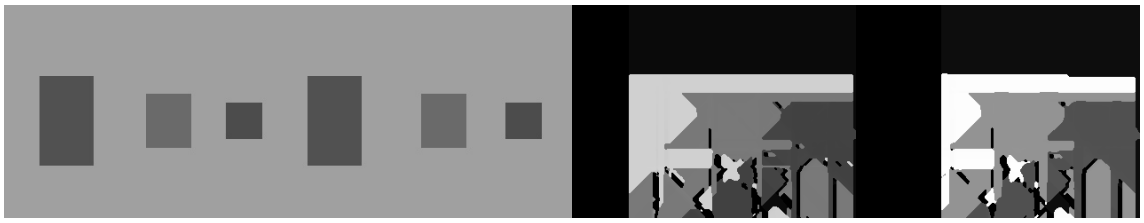


## 6. Task 5 - Stereo depth (2 marks)

StereoSGBM disparity; nearer should look brighter. Parameter effects shown for numDisparities/blockSize. Nearer objects should look brighter (larger disparity). My stereo pair is roughly rectified; a proper rig would

improve alignment.

**Figure 7. Stereo + disparity maps**



## References


- Gonzalez, R. C. & Woods, R. E. Digital Image Processing. (course textbook / lecture notes).
- OpenCV documentation: https://docs.opencv.org/
- Lecture slides for CSC 6302 Computer Vision (UNZA), 2025/26.
- OpenCV camera calibration and solvePnP tutorials.
- Welch & Bishop Kalman filter notes (constant velocity model).


## AI use

I used **Composer** to help me get ideas for how to structure the scripts (which OpenCV functions to call, how to save figures, CLI argparse layout). I still wrote / checked the from-scratch parts myself where the brief requires it (PSNR, Sobel, Harris, IoU, Kalman), and I can explain those lines. The photos in this report are the ones in my `images/` folder. Any numbers in the tables come from running my code.